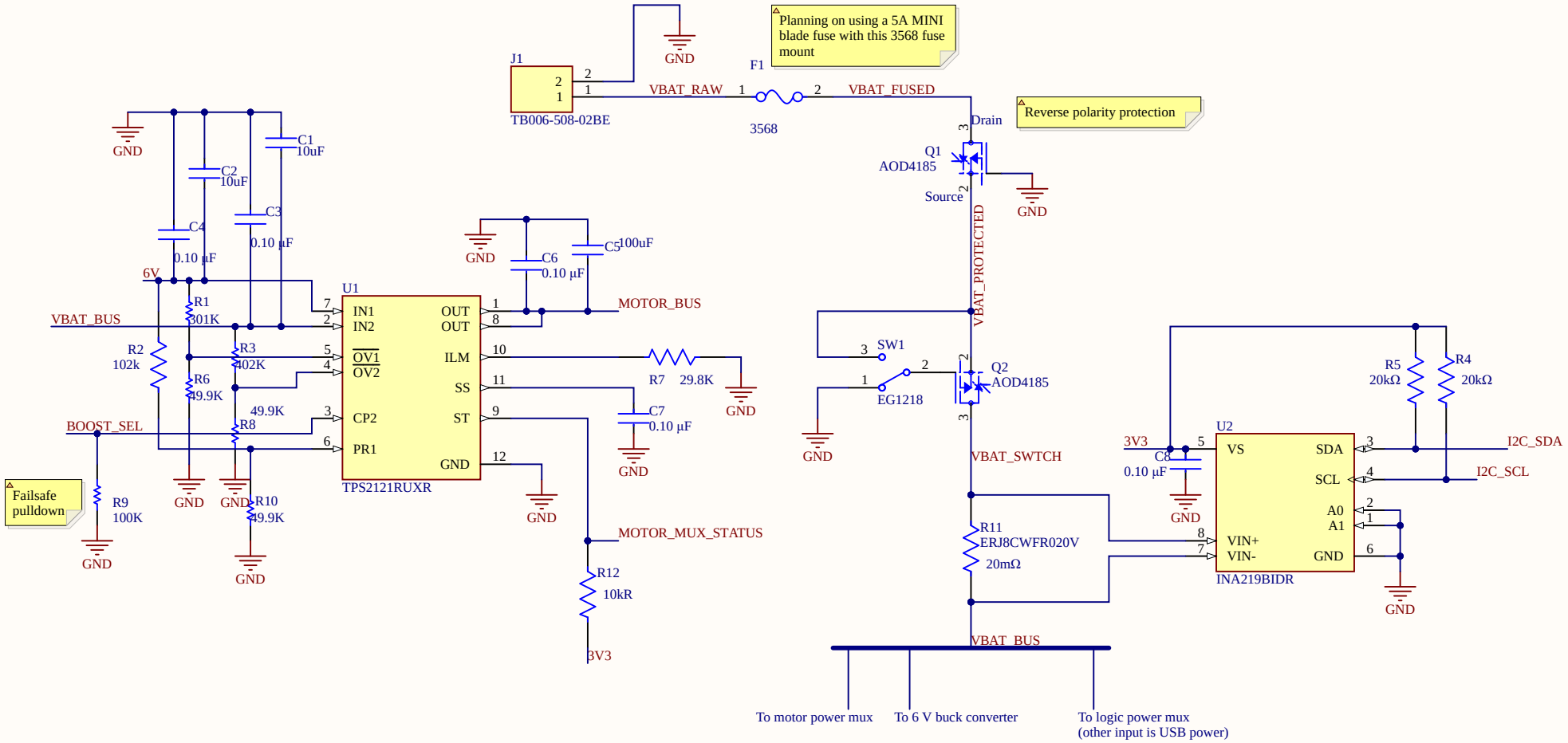
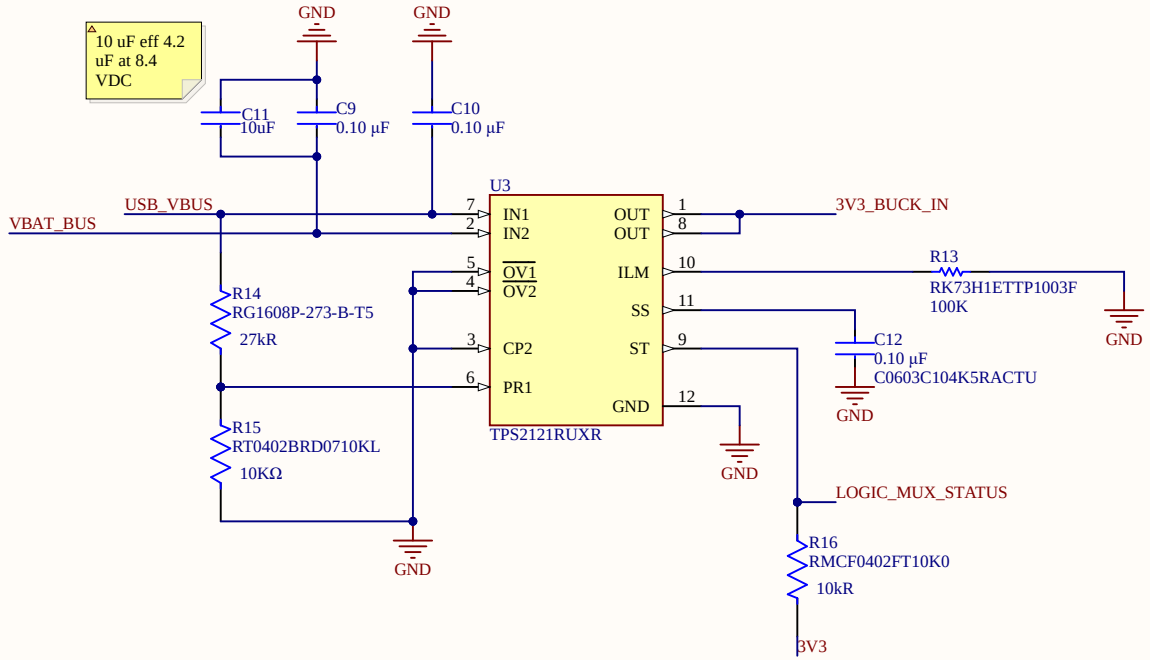
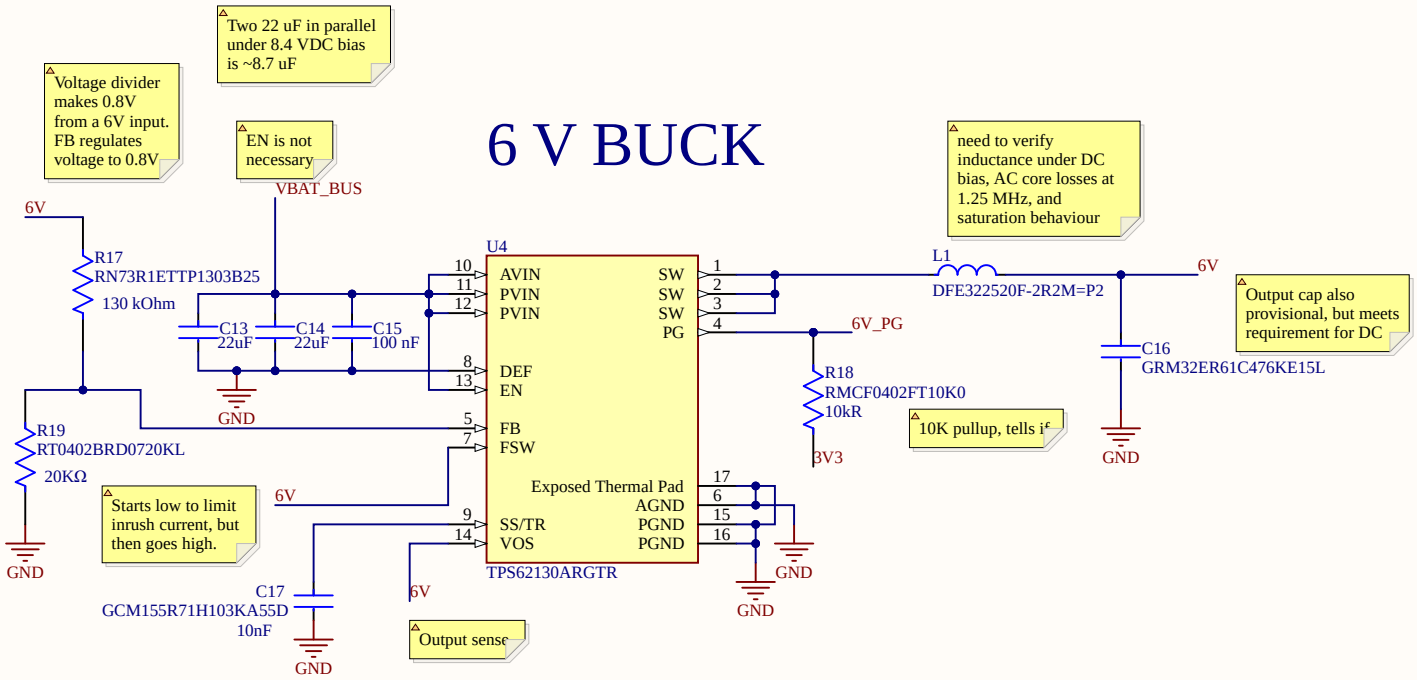


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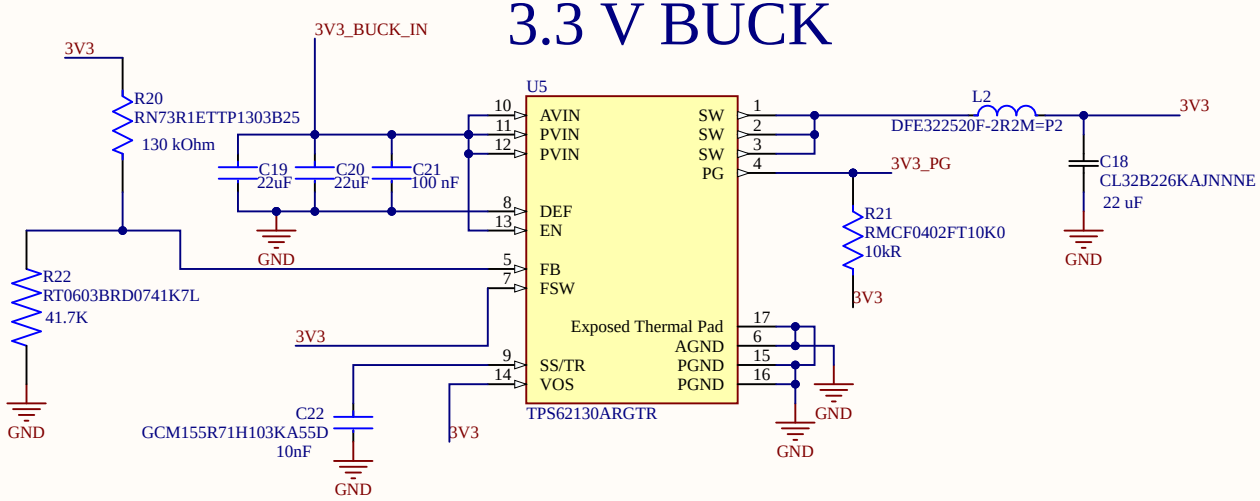


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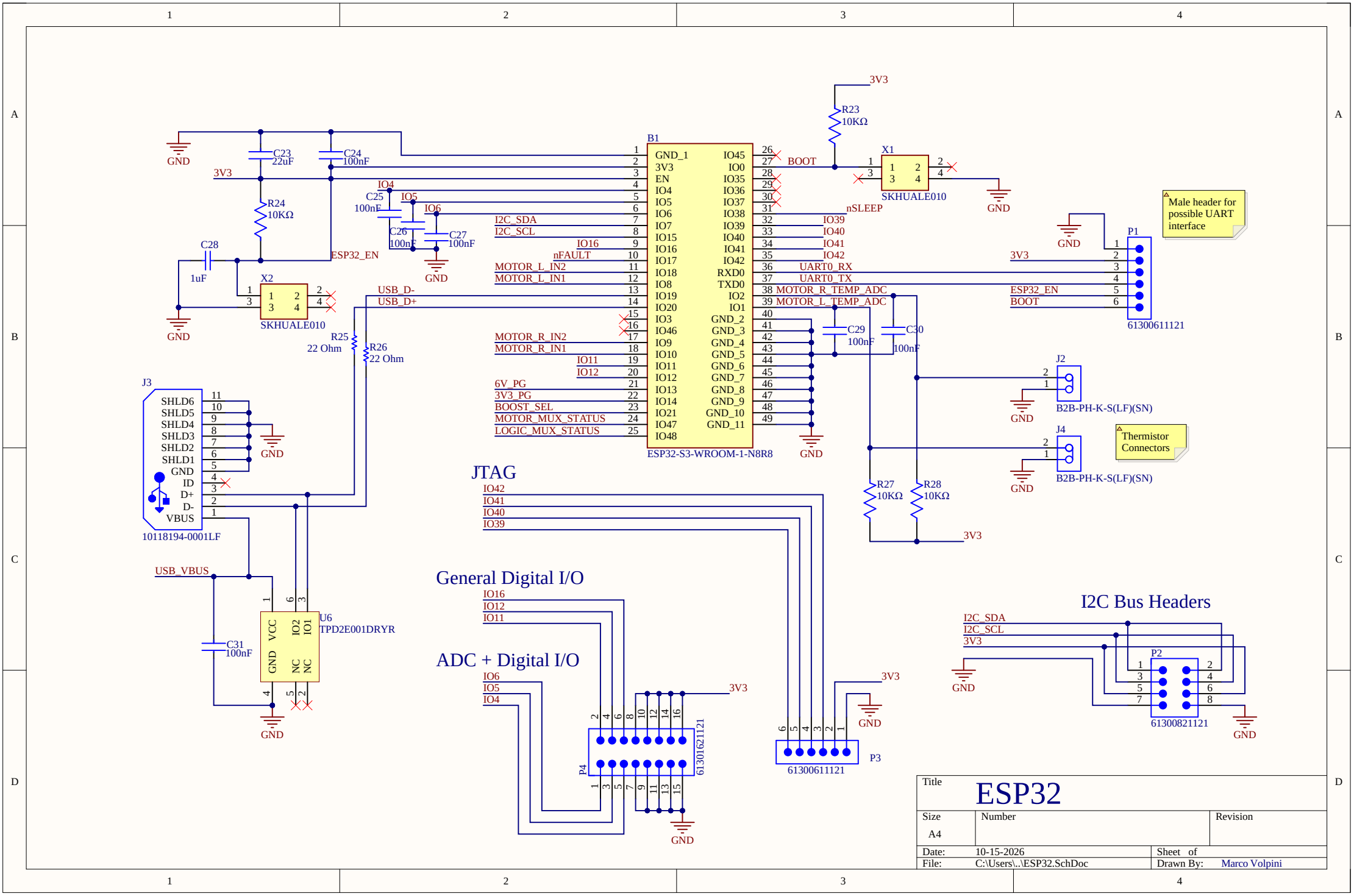
6 V BUCK



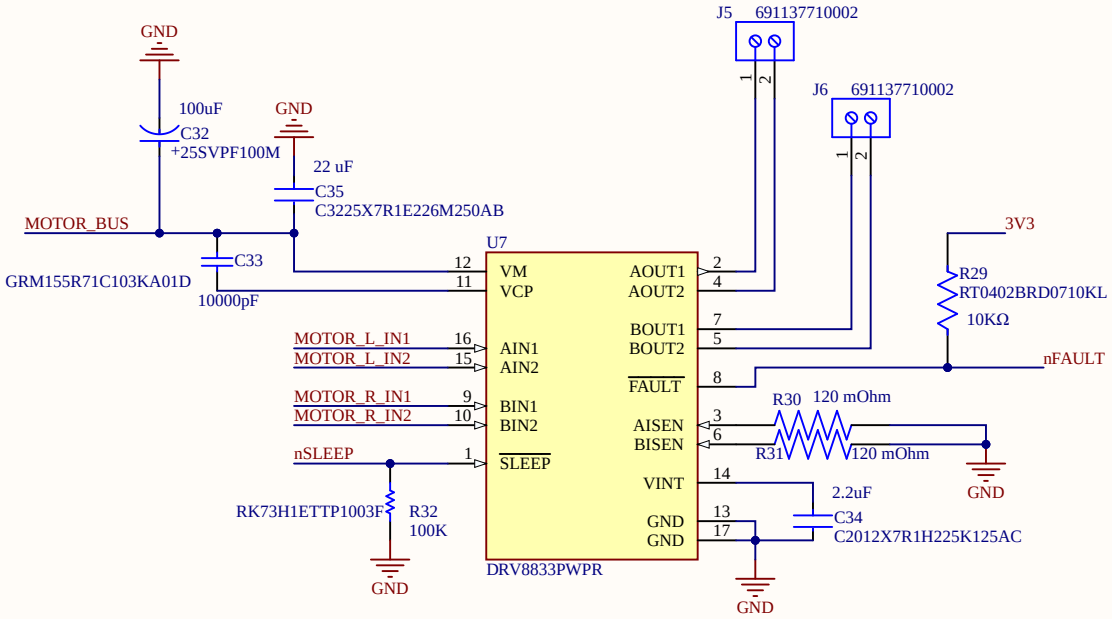
3.3 V BUCK



Title: Buck Converters		
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Title	ESP32	
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File: C:\Users\..\ESP32.SchDoc		Drawn By: Marco Volpini



Title	Motor Driver	
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Date: 10-15-2026		Sheet of
File: C:\Users\..\motor_driver.SchDoc		Drawn By: Marco Volpini

Line #	Name	Description	Designator	Revision ID	Revision State	Revision Status	Quantity	Manufacturer 1	Manufacturer Part Number 1	Manufacturer Lifecycle 1	Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Subtotal 1
	ESP32-S3-WROOM-1-N8R8	WiFi Modules - 802.11 (Engineering Samples Only) SMD Module, ESP32-S3R8 with 8 MB Octal PSRAM Rev. 0 MB Quad SPI flash, PCB Antenna	B1		Not managed		1							
	GRM21BR61H106KE43L	Chip Multilayer Ceramic Capacitors for General Purpose, 0805, 10uF, X5R, 10%, 10%, 50V	C1, C2, C11	CMP-06035-004402-1	New From Design	Up to date	3	Murata	GRM21BR61H106KE43L	Volume Production				
	C0603C104K5RACTU	CAP CER 0.1UF 50V X7R 0603	C3, C4, C6, C7, C8, C9, C10, C12	CMP-2006-03061-2	New From Design	Up to date	8	Yageo/Group	C0603C104K5RACTU	Volume Production	Mouser	80-C0603C104K5R	0.07	0.7
	EMK225ABJ107MA-T	CAP CER 100UF 16V X5R 1210	C5	CMP-2000-05020-2	New From Design	Up to date	1	TAIYO YUDEN	EMK225ABJ107MA-T	Obsolete	Mouser	963-EMK225ABJ107MA-T	2.57	2.57
	CL21A226KOQNNNG	0.1.21 Series 0805 22uF 10V ±10% Tolerance X5R Multilayer Ceramic Chip Capacitor	C13, C14, C19, C20	CMP-13271-000211-1	New From Design	Up to date	4	Samsung Electro-Mechanics	CL21A226KOQNNNG	Not Recommended for New Design	Mouser	187-CL21A226KOQNNNG	0.42	1.68
	CC0402KRX7R7BB104	CAP CER 0.1UF 16V X7R 0402	C15, C21	CMP-03422-000048-1	New From Design	Up to date	2	Yageo/Group	CC0402KRX7R7BB104	Volume Production	Newark	CC0402KRX7R7BB104	0.035	0.05
	GRM21BR61C476KE15L	CAP CER 47UF 16V X5R 1210	C16	CMP-2000-06362-2	Released	Up to date	1	Murata	GRM21BR61C476KE15L	Not Recommended for New Design	DigiKey	490-6539-6-ND	1.43	1.43
	GCM155R71H103KA55D	Ceramic Capacitor for Automotive 10nF ±10% 50VDC X7R 0402 Paper T/R	C17, C22	CMP-2008-03723-3	New From Design	Up to date	2	Murata	GCM155R71H103KA55D	Volume Production	ArrowElectronics	GCM155R71H103KA55D	0.0145	0.145
	CL31B226KAJNNNE	Description	C18		Not managed		1							
	CL31B226KPHNNNE	Cap Ceramic 22uF 10V X7R ±10% SMD 1206 +125°C Embossed T/R	C23	CMP-13271-003516-1	New From Design	Up to date	1	Samsung Electro-Mechanics	CL31B226KPHNNNE	Volume Production	Mouser	187-CL31B226KPHNNNE	0.5	0.5
	CL05B104KP5NNNC	Cap Ceramic 100nF 10V X7R ±10% SMD 0402 +125°C Paper T/R	C24, C25, C26, C27, C29, C30, C31	CMP-13271-001069-1	New From Design	Up to date	7	Samsung Electro-Mechanics	CL05B104KP5NNNC	Volume Production	DigiKey	1276-1002-1-ND	0.025	0.25
	GRM188R61A225KE15D	Chip Capacitor, 1 µF, +/- 10%, 10 V - 55 to 85 degC, 0603 (1608 Metric), RoHS, Tape and Reel	C28	CMP-2000-06336-1	Released	Up to date	1	Murata	GRM188R61A105KA61D	Not Recommended for New Design	ArrowElectronics	GRM188R61A105KA61D	0.108	0.108
	25SMF100M	Conductive Polymer Aluminum Solid Capacitors 0.25 CDV 100 µF ±20% 25V RADIAL SMT B45 6mm	C32	CMP-05428-000915-1	New From Design	Up to date	1	Panasonic	25SMF100M	Volume Production	Arrel	25SMF100M	2.42	2.42
	GRM155R71C103KA01D	Chip Capacitor, 10 nF, 16 V, 10% -55 to 125 degC, 0402 (1005 Metric), RoHS, Tape and Reel	C33	CMP-2000-06332-1	Released	Up to date	1	Murata	GRM155R71C103KA01D	Not Recommended for New Design	ArrowElectronics	GRM155R71C103KA01D	0.0964	0.0964
	C2012X7R1H225K125AC	Multilayer Ceramic Capacitors 2.2µF ±10% 50V X7R 0805	C34	CMP-2000-06040-2	Released	Up to date	1	TDK	C2012X7R1H225K125AC	Volume Production	ArrowElectronics	C2012X7R1H225K125AC	0.4917	0.4917
	C3225X7R1E226M250AB	Multilayer Ceramic Capacitors 22uF ±20% 25V X7R SMD 1210	C35	CMP-08246-001769-2	New From Design	Up to date	1	TDK	C3225X7R1E226M250AB	Volume Production	ArrowElectronics	C3225X7R1E226M250AB	1.26	1.26
	3568		B1	CMP-2000-05464-2	Released	Up to date	1	Keystone	3568	Volume Production	Newark	3568	1.08	1.08
	TB006-508-02BE	2 - 24 Poles, Screw Type, Horizontal, 5.08 Pitch, 26 - 14 AWG, Terminal Block Connector	J1		Not managed		1	Same Sky	TB006-508-02BE	Volume Production	ArrowElectronics	TB006-508-02BE	0.1624	100.04
	B2B-PH-K-S(LF)(SN)	Male Header, Pitch 2 mm, 1 x 2 Position, Height 6 mm, Tail Length 3.4 mm, -25 to 85 degC, RoHS, Bulk	J2, J4	CMP-2000-05162-1	Released	Up to date	2	JST	B2B-PH-K-S(LF)(SN)	Volume Production	Newark	B2B-PH-K-S(LF)(SN)	0.099	0.198
	10118194-0001LF	Micro USB B Type Receptacle, 1.8 A, 5 V, -55 to 85 degC, 5 - Pin SMD, RoHS, Tape and Reel	J3	CMP-2000-05120-1	Released	Up to date	1	Amphenol Communications Solutions	10118194-0001LF	Volume Production	ArrowElectronics	10118194-0001LF	0.439	0.439
	691137710002	Terminal Block, 5MM Pitch, Horizontal, Pressure Clamp, Glow-Wire, Pcb, 2Pins, Gy	J5, J6	CMP-1502-03324-3	Released	Up to date	2	Wurth Elektronik	691137710002	Volume Production	DigiKey	732-10995-ND	0.38	0.76
	DFE322520F-2R2M-P2	DFE322520F-2R2M-P2 Inductor Power Shielded Wirewound 2.2uH 20% 1MHz Metal 3.1A/0.04Ohm SMD 1212 T/R	L1, L2	CMP-00260-130381681-1	New From Design	Up to date	2	Murata	DFE322520F-2R2M-P2	Volume Production	ArrowElectronics	DFE322520F-2R2M-P2	0.2692	0.5384
	61300611121	THT Vertical Pin Header WR-PHD, Pitch 2.54 mm, Single Row, 6 pins	P1, P3	CMP-1502-01070-4	Released	Up to date	2	Wurth Elektronik	61300611121	Volume Production	DigiKey	732-5319-ND	0.33	0.66
	61300821121	THT Vertical Pin Header WR-PHD, Pitch 2.54 mm, Dual Row, 8 pins	P2	CMP-1502-01078-4	Released	Up to date	1	Wurth Elektronik	61300821121	Volume Production	DigiKey	732-5296-ND	0.67	0.67
	61301621121	THT Vertical Pin Header WR-PHD, Pitch 2.54 mm, Dual Row, 16 pins	P4	CMP-1502-01097-4	Released	Up to date	1	Wurth Elektronik	61301621121	Volume Production	DigiKey	732-5299-ND	1	1
	AOD4185	P-Channel 40V 40A (Tc) 2.5W (Ta), 62.5W (Tc) Surface Mount TO-252, (D-Pak)	Q1, Q2		Not managed		2							
	RK73H1ETTP5013F	2.5kΩ ±1% 0.125W, 1/8W Chip Resistor 0805 (2012 Metric) Moisture Resistant, Thin Film	R1	CMP-2002-03895-1	Released	Up to date	1	KOA Speer	RK73H1ETTP5013F	Volume Production	DigiKey	2019-RK73H1ETTP5013FCT-ND	0.1	0.1
	RMCF0603FG102K	102kΩ ±1% 0.1W, 1/10W 0603 Thick Film Chip Resistor AEC-Q200 compliant	R2	CMP-28527-000784-1	New From Design	Up to date	1	Stackpole Electronics	RMCF0603FG102K	Volume Production	DigiKey	738-RMCF0603FG102KCT-ND	0.1	0.1
	RC0805FR-0740ZKL		R3	CMP-2001-011336-1	Released	Up to date	1	Yageo/Group	RC0805FR-0740ZKL	Volume Production	ArrowElectronics	RC0805FR-0740ZKL	0.048	0.048
	RC0402FR-0720KL		R4, R5	CMP-2002-07501-2	Released	Up to date	2	Yageo/Group	RC0402FR-0720KL	Volume Production	ArrowElectronics	RC0402FR-0720KL	0.0097	0.093
	TNPU0603499KB2BEN00		R6, R8, R10	CMP-2000-03603-1	Released	Up to date	3	Vishay	TNPU0603499KB2BEN00	Volume Production	DigiKey	TNPU0499KB4CT-ND	2.1	6.3
	RN73R2ATTD2982B50	29.8kΩ ±0.1% 0.125W, 1/8W Chip Resistor 0805 (2012 Metric) Moisture Resistant, Thin Film	R7	CMP-04015-062765-1	New From Design	Up to date	1	KOA Speer	RN73R2ATTD2982B50	Volume Production	DigiKey	2019-RN73R2ATTD2982B50CT-ND	0.13	0.13
	RK73H1ETTP5003F		R9, R13, R32	CMP-2002-04173-1	Released	Up to date	3	KOA Speer	RK73H1ETTP5003F	Volume Production	ArrowElectronics	RK73H1ETTP5003F	0.0155	0.155
	ERJ8CSMFR025V		R11	CMP-2003-007533-2	Released	Up to date	1	Panasonic	ERJ8CSMFR025V	Volume Production	ArrowElectronics	ERJ8CSMFR025V	0.4245	0.4245
	RMCF0402FT10K0	10kΩ ±1% 0.063W, 1/16W 0402 Thick Film Chip Resistor AEC-Q200 compliant	R12, R16, R18, R21	CMP-28527-000128-1	New From Design	Up to date	4	Stackpole Electronics	RMCF0402FT10K0	Volume Production	Mouser	708-RMCF0402FT10K0	0.012	0.12
	RG1608P-273-B-T5	Automotive Metal Thin Film Chip Resistor, 0603, 27kΩ, 0.1%, 25ppm/°C, 0.1W, 100V	R14	CMP-2000-02157-2	Released	Up to date	1	Susumu	RG1608P-273-B-T5	Volume Production	ArrowElectronics	RG1608P-273-B-T5	0.0964	0.0964
	RT0402BRD0710KL	9.53kΩ ±0.1% 0.1W, 1/10W 0402	R15, R29	CMP-2002-02915-2	New From Design	Up to date	2	Yageo/Group	RT0402BRD0710KL	Volume Production	ArrowElectronics	RT0402BRD0710KL	0.0964	0.1928
	RN73R1ETTP13030B25	130kΩ ±0.1% 0.063W, 1/16W Chip Resistor 0402 (1005 Metric) Moisture Resistant, Thin Film	R17, R20	CMP-04015-027314-1	New From Design	Up to date	2	KOA Speer	RN73R1ETTP13030B25	Volume Production	DigiKey	2019-RN73R1ETTP13030B25CT-ND	0.1	0.2
	RT0402BRD0720KL	953.5kΩ ±0.1% 0.1W, 1/10W 0402	R19	CMP-2002-02936-2	New From Design	Up to date	1	Yageo/Group	RT0402BRD0720KL	Volume Production	ArrowElectronics	RT0402BRD0720KL	0.0964	0.0964
	RT0603BRD0741KL	41.7kΩ ±0.1% 0.1W, 1/10W Chip Resistor 0603 (1608 Metric) Thin Film	R22		Not managed		1							
	RT0603BRD0710KL	10kΩ ±0.1% 0.1W, 1/10W 0603	R23, R24, R27, R28	CMP-03412-000619-1	New From Design	Up to date	4	Yageo/Group	RT0603BRD0710KL	Volume Production	ArrowElectronics	RT0603BRD0710KL	0.0964	0.3856
	CRCW040222R0FKED		R25, R26	CMP-2002-08382-1	Released	Up to date	2	Vishay	CRCW040222R0FKED	Volume Production	ArrowElectronics	CRCW040222R0FKED	0.0024	0.0048
	ERJ8ECPR12V		R30, R31		Not managed		2				Newark	ERJ8ECPR12V		
	EG1218	Slide Switch, SPDT, 6 V, -20 to 70 degC, Thru Hole, RoHS	SW1	CMP-1696-000002-1	Released	Up to date	1	C&K	EG1218	Volume Production	DigiKey	EG1903-ND	0.72	0.72
	TPS2121RUNR	PMIC - Power Distribution Switches, Load Drivers USB Switch IC 2:1 Multiplexer 10-UQFN (1.8x1.4)	U1, U3	CMP-04910-000009-1	New From Design	Up to date	2	Texas Instruments	TPS2121RUNR	Volume Production	DigiKey	296-53410-6-ND	2.44	4.88
	INA219BIDR	IC SENSOR PWR/CURR/I2C 8SOIC	U2	CMP-0704-000035-3	Released	Up to date	1	Texas Instruments	INA219BIDR	Volume Production	ArrowElectronics	INA219BIDR	1.88	1.88
	TPS62130ARGTR	IC REG BUCK ADJUSTABLE 3A 16QFN	U4, U5	CMP-04918-000067-1	New From Design	Up to date	2	Texas Instruments	TPS62130ARGTR	Volume Production	Newark	TPS62130ARGTR	1.99	3.98
	TPD2E001DRRR	Low-Capacitance +/- 15-kV ESD-Protection Array for High-Speed Data Interfaces, 2 Channels, -40 to +85 degC, 6-pin SON (DRR), Green (RoHS & no Sb/Br)	U6	CMP-0326-000025-3	Released	Up to date	1	Texas Instruments	TPD2E001DRRR	Volume Production	ArrowElectronics	TPD2E001DRRR	0.3021	0.3021
	DRA8033PWPR	IC MICROSENSR PRT 28TSSOP	U7	CMP-1529-00004-3	Released	Up to date	1	Texas Instruments	DRA8033PWPR	Volume Production	Mouser	595-DRA8033PWPR	3.03	3.03
	SHHLALE010	Tactile Switches 6.0x6.0x2.5mm 160gf	K1, K2		Not managed		2	ALPS	SHHLALE010	Volume Production	Newark	SHHLALE010	0.3873	3.87